# ALINA (AMELIE) KAPANOVA

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## EDUCATION

**Cornell University**, Cornell Tech, New York, United States — Graduation date: 5/2027
***Master of Science in Information Systems*** | GPA: 3.94
**Relevant courses**: Applied Machine Learning, Privacy, ML Engineering, Algorithms, Practical Applications of ML, Startup Studio.

**New York University,** New York, United States — Graduation date: 9/2022
***Bachelor of Science in Computer Science*** | GPA: 3.84/4.0, CS GPA 3.97/4.0
**Honors**: Magna Cum Laude (top 10%) honors, Founders Day Award, NYU merit scholarship (42000 USD).

## PROFESSIONAL EXPERIENCE

**Wisense Lab, Cornell University, Cornell Tech, Research Assistant** — 8/2026 - Present
- Developing machine-learning technologies for real-world applications in mobile health, ubiquitous computing, user interfaces, and IoT.

**Anote, MLE Intern**, New York, United States — 8/2026 - Present
- Conducted research on agentic AI, multi-agent systems, and LLM evaluation, working on paper conference submissions

**Anote, AI Research Fellow & MLE Intern**, New York, United States — 5/2025- 8/2025, 6/2026-8/2026
- Conducted research on agentic AI, multi-agent systems, and LLM evaluation, developing **MetaRouting** benchmarks for intelligent tool and agent selection and evaluating routing strategies across performance, cost, latency, prepared for submission to **AAAI**, EMNLP.
- Developed a synthetic video generation algorithm using Python, Replicate API, researched current methods and approaches to synthetic training data generation, rehired for fellowship in 2026.

**Cornell University, Cornell Tech Research Assistant**, New York, United States — 9/2023-12/2024, 8/2024-5/2025
- Assisting at the Urban Tech Hub in research at the intersection of Urban and CS. Wrote a Python script that evaluated LLM model performance based on Stanford's HELM study, rehired for second academic year.
- Developed an NLP tool to analyze climate strategies aligned with the Toronto Resilience Strategy. Applied LDA topic modeling and bigram analysis using Python (NLTK, Scikit-learn, Gensim) to extract themes on climate adaptation.

**Amazon.com , Software Development Engineer**, Dublin, Ireland — 6/2022- 2/2023
- Created, deployed a Java package to validate production environments for crucial data pipelines to reduce tickets by 35%. Deployed MemDS to 3 new commercial regions, interacting across the USA, India offices, resolving tickets, making and verifying code changes

**Agritecture Consulting, Software Engineer Intern**, Remote — 3/2020- 5/2020
- Built Daily Light integral Calculator (DLI) web app using JavaScript, Google's Geocoding, Maps API. The app calculates average monthly DLI values for any location, retrieves geocoding info, computes DLI in 6 steps. The app was hosted on company website.

## PROJECTS

**Named Entity Recognition ML pipeline, Opengulf lab at Stanford and NYU AD - Remote**. — 5/2021 – 2/2022
- Implemented a custom **Named Entity Recognition (NER) model in Python** based on spaCy's LSTM architecture, pretrained pipelines, tags such as "tribe" to capture historical names. The model addresses the underrepresentation of historical, non-English entities in existing NER systems. The work was published at the **Digital Humanities Conference 2022 (Sweden)**.

**Research Assistant, Secure Systems lab at NYU Tandon's School of Engineering** — 01/2021 – 03/2022
- Assisted in Crash Simulator project, a tool identifying vulnerabilities in product designs, Conducted a network performance study by recording system calls with strace in C, wrote unit tests (Python) to increase test coverage for posix-omni-parser from 71% to 95%.

## SKILLS

**Technical skills:** Machine Learning, AI, Software Engineering, Full-stack development, Unit Testing, Data Analysis, Deep Learning, LLMs, Named Entity Recognition, MongoDB, ReactJS, SpaCy, Pytorch, NodeJS, ExpressJS, Java, Python, JavaScript, SQL, C, C++.

## PUBLICATIONS

**- Cross-lingual transfer, especially through Kannada transliteration, improves Tulu legal comprehension, while RAG can introduce fact substitution and hallucinations. Accepted to an EMNLP 2026 workshop. Paper.**
**- This paper introduces an executable benchmark and budget-aware meta-router for agentic workflows, achieving 100% held-out task success at 43% lower cost than static routing while identifying lexical generalization as the main limitation. Paper.**
**- Published at Digital Humanities 2022, this paper develops custom spaCy-based NER models to better identify transliterated Arabic and Farsi entities in multilingual historical collections from the Persian Gulf. Paper.**

## HONORS

**NYU Undergraduate merit scholarship (42k), New York University Magna Cum Laude Latin Honors in CS** (top 10%), 2022
**NYU Tandon's Undergraduate Summer Research Program, NYU AD Visiting Summer Research Program**, 2021
**Grace Hopper EMEA** scholar by Anita Borg (2021), **Grace Hopper US Scholar** by AccessStem (University of Washington, 2020)

## EXTRACURRICULARS

**Cornell Tech, Student Ambassador, Urban Tech rep,** 01/2025 - 04/2025, support incoming students, tours, guide through questions
**Cornell Tech Dom Kultury: ML & Art Club, President; Community Service Club, Communications Chair.**